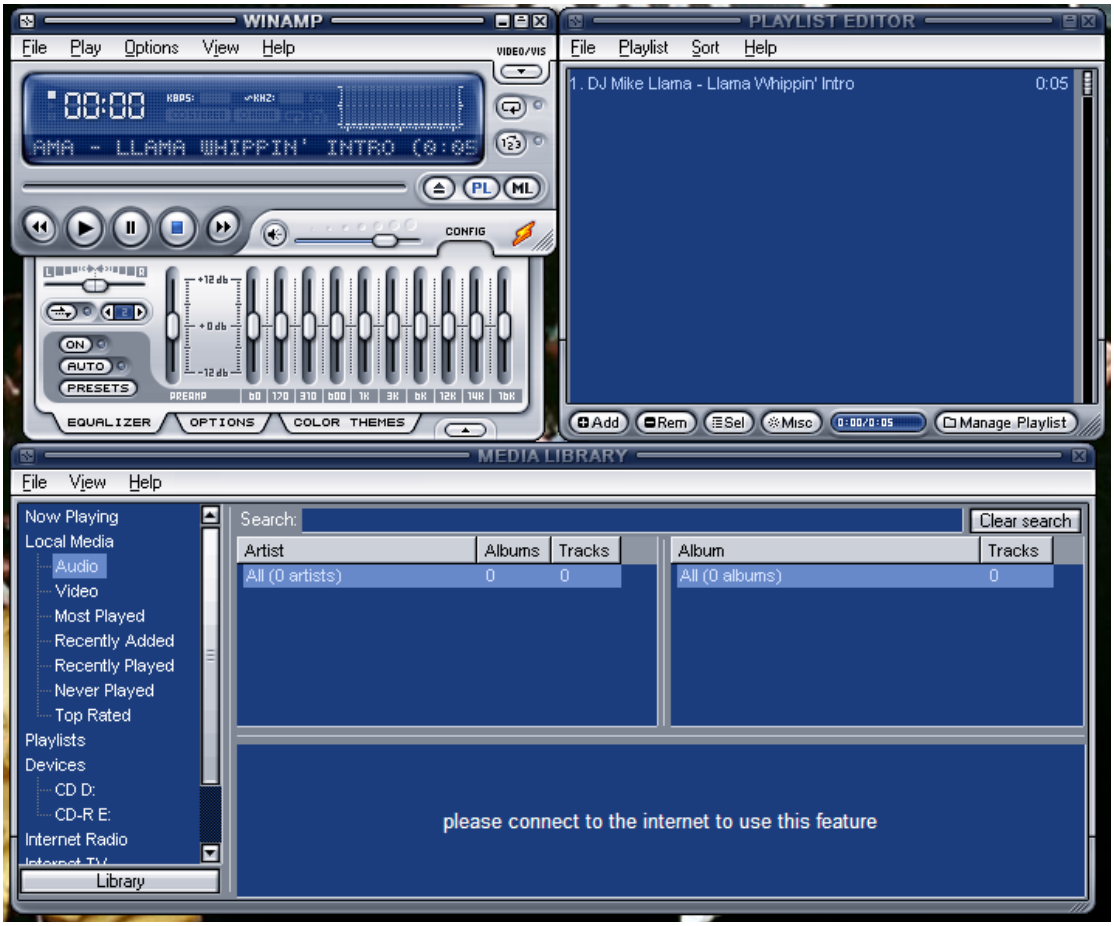
Alexandria University
Faculty of Engineering
Computer and communication program

جامعة الاسكندرية
كلية الهندسة

DSP
6th term
Instructors:Prof. Hassan El Ragal Prof. Ahmed Eltrass, and Prof.Mona lofty

Final Project

It is required to develop audio equalizer using MATLAB. One common Audio equalizer is the winamp program as shown in figure 1. The equalizer function is to vary the gain of each specific band as the user prefers. [i.e., if the user likes base, he will increase the gain of low frequencies].



Assignment Procedure:

Inputs to program:

User should input the following:
1) The wave file name.
2) Gain of each of the frequency bands in db.
3) Type of filters used (IIR-FIR)
4) Output sample rate.
You may make use of the following functions: input – menu – uigetfile.

Method:

1) Develop the frequency band filters in the following bands
(0- 200 Hz) - (200- 400 Hz) - (400- 800 Hz) -(800- 1200 Hz) -(1200Hz- 3 KHz) - (3- 6 KHz) - (6-12 KHz) - (12-15 KHz) - (15-20 KHz)
2) Analyze and export these files (Gain, phase, impulse and step response, order and poles/zeros).
3) Filter the wave file using the filters developed in step 1.
4) Draw the output signals in Time and frequency domains.
5) Amplify the output signals using the user defined gain.
6) Add the amplified - output signals in time domain to form composite signal.
7) Draw and compare the composite signal with the original signal. (in time and frequency)
8) Play and save the output wave signal. (you can use wavwrite-sound)

Outputs/Requirements of project:
You should submit ONE PDF file contains the followings:

1) Copy of your code.

2) Different sample runs of code including the following cases:
a) If design is using IIR filters.
b) If design is using FIR filters.
c) Output signal in case if output sample rate is multiplied by three or decreasing it to half

3) The analysis of each filter and the exported outputs.

4) All figures of signals in time and frequency domain.

*User interface to input the data to M- file even if as command window lines (GUI).